Vector Calculus, Linear Algebra and Differential Forms: A Unified Approach - Solutions

June 26, 2026

Contents

Chapter 0

Preliminaries

0.2 Quantifiers and Negation

Exercise 0.2.1 Negate the following statements:

a. Every prime number such that if you divide it by 4 you have a remainder of 1 is the sum of two squares.

b. For all $x \in \mathbb{R}$ and for all $\epsilon > 0$, there exists $\delta > 0$ such that for all $y \in \mathbb{R}$, if $|y-x| < \delta$, then $|y^2-x^2| < \epsilon$.

c. For all $\epsilon > 0$, there exists $\delta > 0$ such that for all $x, y \in \mathbb{R}$, if $|y-x| < \delta$, then $|y^2-x^2| < \epsilon$.

SOLUTION

a. There exists a prime number in the form of $4n+1$, for $n \in \mathbb{N}_0$ that cannot be expressed as the sum of two squares.

b. There exists $x \in \mathbb{R}$ and $\epsilon > 0$ such that for all $\delta > 0$, there exists $y \in \mathbb{R}$ with $|y-x| < \delta$ but $|y^2-x^2| \geq \epsilon$.

c. There exists $\epsilon > 0$ such that for all $\delta > 0$, there exists $x, y \in \mathbb{R}$ with $|y-x| < \delta$ but $|y^2-x^2| \geq \epsilon$.

Exercise 0.2.2 Explain why one of these statements is true and the other is false:

$$(\forall \text{ man } M)(\exists \text{ woman } W) \mid W \text{ is the mother of M}$$
$$(\exists \text{ woman } W)(\forall \text{ man } M) \mid W \text{ is the mother of M}$$

SOLUTION

The difference between the two statements lies in the quantifier. Obviously the first statement is true while the second is false.

The first statement states that for every man, we can find a woman that is the man's father. This is inherently true as every man must have a mother.

On the other hand, the second statement states that we can find a women that is the mother of every man, which is obviously not possible.

0.3 Set Theory

Exercise 0.3.1 Let E be a set, with subsets $A \subset E$ and $B \subset E$, and let $*$ be the operation $A * B = (E - A) \cap (E - B)$. Express the sets

a. $A \cup B$ b. $A \cap B$ c. $E - A$

using A, B and $*$.

SOLUTION

Before solving this problem, consider the special case of the operator $*$:

$$X * X = (E - X) \cap (E - X) = X^c \cap X^c = X^c$$

a. We can first simpify the operator with $A * B = A^c \cap B^c = (A \cup B)^c$. Using the special case, we get

$$(A * B) * (A * B) = (A \cup B)^c * (A \cup B)^c = A \cup B$$

b. Working backwards, we get

$$A \cap B = (A^c \cup B^c)^c = ((A * A) \cup (B * B))^c = (A * A) * (B * B)$$

c. This case is the equivilant to the special case above, hencec $E - A = A^c = A * A$.

0.4 Functions

Exercise 0.4.1 Are the following true functions? That is, are they both everywhere defined and well defined?

a. "The aunt of", from people to people.

b. $f(x) = \frac{1}{x}$, from real numbers to real numbers.

c. "The capital of", from countries to cities (careful - at least one country, Bolivia, has to capitals.)

SOLUTION

a. Considering the cases where and individual's parents not having any sisters, the individual does not have an aunt, making it not everywhere defined. Considering the cases where an individual has multiple aunts, the function is not well defined.

b. The function is not everywhere defined on real numbers as when $x = 0$, $f(x)$ is undefined. The function however is wel defined as if $f(x) = f(y)$, then $x = y$.

c. The function is everywhere defined as every country has a capital. However, it is not well defined as some countries, such as Bolivia, has two capitals.

Exercise 0.4.2 a. Make up a nonmathematical function that is onto but not one to one.

b. Make up a mathematical function that is onto but not one to one.

SOLUTION

a. The suit of a poker card is onto but not one to one; every card has a suit but more that one card has the same suit.

b. $f : \mathbb{R} \to \mathbb{R} \quad f(x) = x^2$

Exercise 0.4.3 a. Make up a nonmathematical function that is bijective (onto and one to one).

b. Make up a mathematical function that is bijective.

SOLUTION

a. The thumbprint of a person is bijective as everyone has a thumbprint that is unique.

b. $f : \mathbb{R} \to \mathbb{R} \quad f(x) = x$.

Exercise 0.4.4 a. Make up a nonmathematical function that is one to one but not onto.

b. Make up a mathematical function that is one to one but not onto.

SOLUTION

a. The function mapping credit card to credit card numbers is one to one but not onto; as each card number is unique but not all card numbers have been used.

b. $f : \mathbb{R} \to \mathbb{R} \quad f(x) = e^x$.

Exercise 0.4.5 Given the functions $f : A \to B$, $g : B \to C$, $h : A \to C$, and $k : C \to A$, which of the following compositions are well defined? For those that are, give the domain and codomain of each.

a. $f \circ g$	b. $g \circ f$	c. $h \circ f$	d. $g \circ h$	e. $k \circ h$	f. $h \circ g$
g. $h \circ k$	h. $k \circ g$	i. $k \circ f$	j. $f \circ h$	k. $f \circ k$	l. $g \circ k$

SOLUTION

a. Not well defined.	b. $g \circ f : A \to C$	c. Not well defined.	d. Not well defined.
e. $k \circ h : A \to A$	f. Not well defined.	g. $h \circ k : C \to C$	h. $k \circ g : B \to A$
i. Not well defined.	j. Not well defined.	k. $f \circ k : C \to B$	l. Not well defined.

Exercise 0.4.6 Prove Proposition 0.4.11. **Inverse image of intersection, union**

a. The inverse image of an intersection equals the intersection of the inverse images:

$$f^{-1}(A \cap B) = f^{-1}(A) \cap f^{-1}(B)$$

b. The inverse image of a union equals the union of the inverse images:

$$f^{-1}(A \cup B) = f^{-1}(A) \cup f^{-1}(B)$$

SOLUTION

a. We will show this by inclusion both ways:

   i. Let $y \in A \cap B$ with $x = f^{-1}(y) \in f^{-1}(A \cap B)$. Then $y \in A$ and $y \in B$, implying $x \in f^{-1}(A)$ and $x \in f^{-1}(B)$, thus $x \in f^{-1}(A) \cap f^{-1}(B)$.

   ii. The inverse is similar. $x \in f^{-1}(A) \cap f^{-1}(B)$, then $x \in f^{-1}(A)$ and $x \in f^{-1}(B)$, so $y \in A$ and $y \in B$. This shows that $y \in A \cap B$, hence, $x \in f^{-1}(A \cap B)$.

b. Similarly, by inclusion both ways:

   i. Let $y \in A \cup B$ with $x = f^{-1}(y) \in f^{-1}(A \cup B)$. Then $y \in A$ or $y \in B$, implying $x \in f^{-1}(A)$ or $x \in f^{-1}(B)$, thus $x \in f^{-1}(A) \cup f^{-1}(B)$.

   ii. The inverse is similar. $x \in f^{-1}(A) \cup f^{-1}(B)$, then $x \in f^{-1}(A)$ or $x \in f^{-1}(B)$, so $y \in A$ or $y \in B$. This shows that $y \in A \cup B$, hence, $x \in f^{-1}(A \cup B)$.

Exercise 0.4.7 Evaluate $(f \circ g \circ h)(x)$ at $x = a$ for the following.

a. $f(x) = x^2 - 1, \quad g(x) = 3x, \quad h(x) = -x + 2, \quad$ for $a = 3$.

b. $f(x) = x^2, \quad g(x) = x - 3, \quad h(x) = x - 3, \quad$ for $a = 1$

SOLUTION

a.

$$\begin{aligned}(f \circ g \circ h)(3) &= f(g(h(3))) \\ &= f(g(-3+2)) = f(g(-1)) \\ &= f(3(-1)) = f(-3) \\ &= (-3)^2 - 1 = 8\end{aligned}$$

b.

$$\begin{aligned}(f \circ g \circ h)(1) &= f(g(h(1))) \\ &= f(g(1-3)) = f(g(-2)) \\ &= f(-2-3) = f(-5) \\ &= (-5)^2 = 25\end{aligned}$$

Exercise 0.4.8 a. Prove Proposition 0.4.16. **(Composition of onto functions).** Let the functions $f : B \to C$ and $g : A \to B$ be onto. Then the composition $(f \circ g)$ is onto.

b. Prove Proposition 0.4.17. **(Composition of one to one functions).** Let $f : B \to C$ and $g : A \to B$ be one to one. Then the composition $(f \circ g)$ is one to one.

SOLUTION

a. Let $c \in C$. Since f is onto, there exists some $b \in B$ such that $f(b) = c$. Since g is onto, there exists some $a \in A$ such that $g(a) = b$. Hence $(f \circ g)(a) = f(g(a)) = f(b) = c$. Since every $c \in C$ has a preimage under $f \circ g$, the composition $f \circ g$ is onto.

b. Let $a_1, a_2 \in A$ and suppose $(f \circ g)(a_1) = (f \circ g)(a_2)$. Then $f(g(a_1)) = f(g(a_2))$. Since f is one to one, $g(a_1) = g(a_2)$. Since g is one to one, $a_1 = a_2$. Therefore $f \circ g$ is one to one.

Exercise 0.4.9 What is the natural domain of $\sqrt{\dfrac{x}{y}}$?

SOLUTION

For $\sqrt{\frac{x}{y}}$ to be defined, $\frac{x}{y} \geq 0$. Hence the natural domain is $\{(x, y) \in \mathbb{R}^2 : y \neq 0, \frac{x}{y} \geq 0\}$.

Exercise 0.4.10 What is the natural domain of

a. $\ln \circ \ln$? b. $\ln \circ \ln \circ \ln$? c. ln composed with itself n times?

SOLUTION

a. The domain of $\ln x$ is $x > 0$. Hence the domain of $\ln \circ \ln$ is when $\ln x > 0$ which is $\{x \in \mathbb{R} : x > 1\}$.

b. Similarly, the natural domain of $\ln \circ \ln \circ \ln$ is $\{x \in \mathbb{R} : x > e\}$.

c. The natural domain of ln composed with itself n times is $\{x \in \mathbb{R} : x > e \uparrow\uparrow (n - 2)\}$.

Exercise 0.4.11 What subset of $\mathbb{R}$ is the natural domain of the function $(1 + x)^{1/x}$?

SOLUTION

For $(1 + x)^{1/x}$ to be defined, we must have the following restrictions, $(1 + x) > 0$; to avoid fractional powers of negative numbers, and $x \neq 0$; as we cannot divide by zero. Hence, the natural domain is $(1, \infty) \setminus \{0\}$.

Exercise 0.4.12 The function $f(x) = x^2$ from real numbers to real nonnegative numbers is onto but not one to one.

a. Can you make it one to one by changing its domain? By changing its codomain?

b. Can you make it not onto by changing its domain? Its codomain?

SOLUTION

a. Yes.

   i. Change the domain to $[0, \infty)$. Then $f(x) = x^2$ is one to one and remains onto $\mathbb{R}_{\geq 0}$.
   ii. Changing the codomain to $\{0\}$, makes $f(x) = x^2$ trivially one to one

b. Yes.

   i. Change the domain to $(0, \infty)$. Then 0 is no longer attained, so f is not onto $\mathbb{R}_{\geq 0}$.
   ii. Change the codomain to $\mathbb{R}$. Then negative real numbers are not attained, so f is not onto.